



05506026

Digital Image Processing

Witchaya Towongpaichayont
Computer Science, KMITL

CONTENTS



1

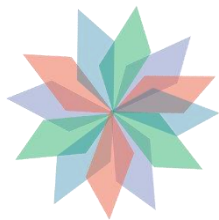
Frequency Domain Processing



FREQUENCY DOMAIN PROCESSING



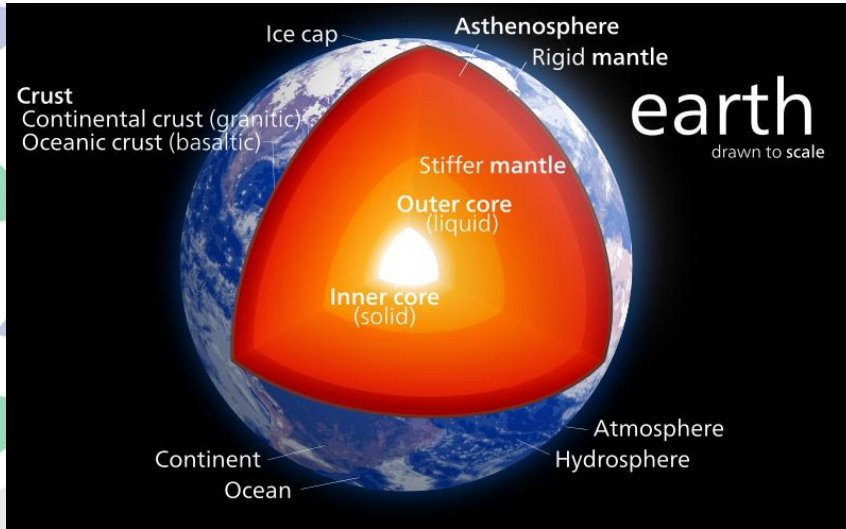
Frequency Domain



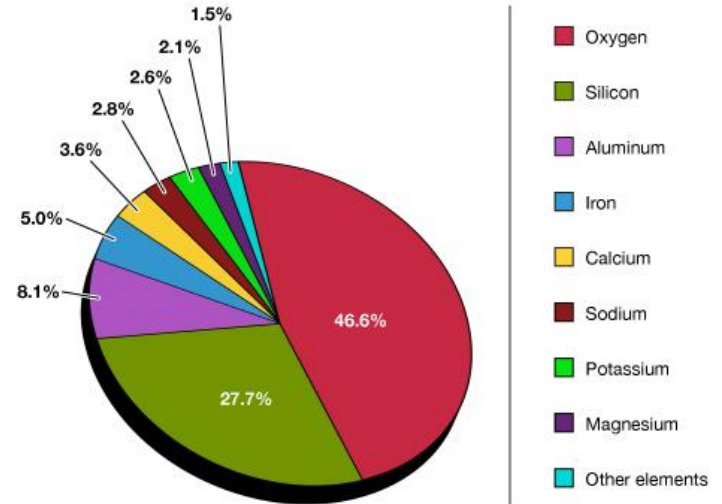
Frequency Domain

- ▶ Frequency Domain is another form of image, apart of spatial domain (matrix of pixels) we have been familiar with
- ▶ Frequency Domain is the mathematical function of images
- ▶ Using Frequency Domain is easier way to deal with some tasks which is difficult to be executed using spatial domain
- ▶ An image in spatial domain can be converted to frequency domain with Fourier Transforms. The image will be displayed in form of Fourier Series

Frequency Domain



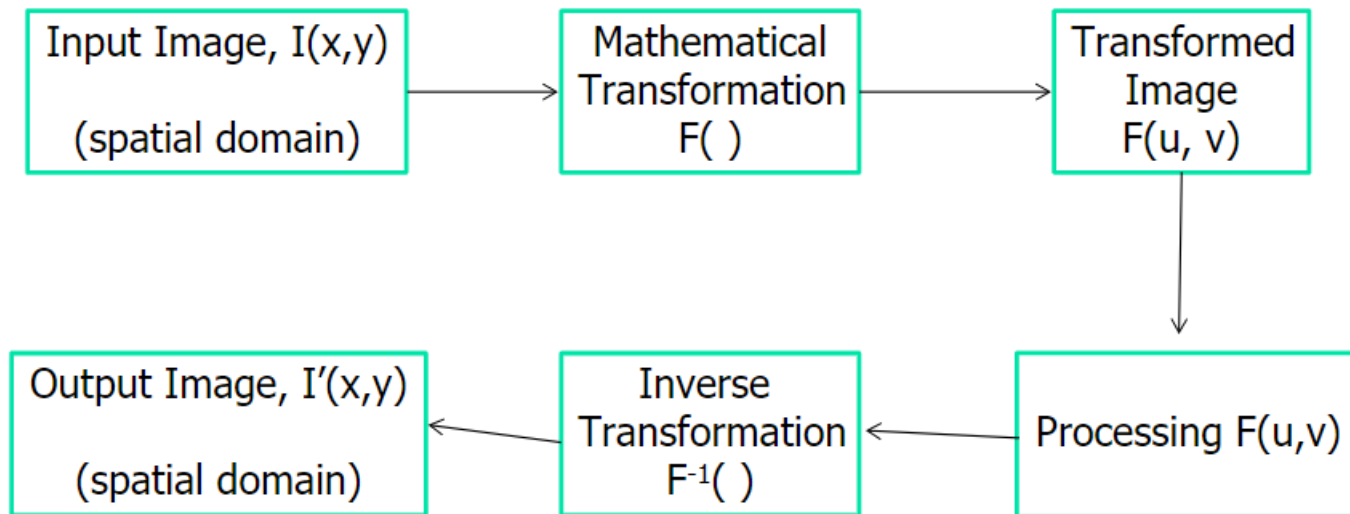
Composition of Earth's Crust



Source: United States Geological Survey

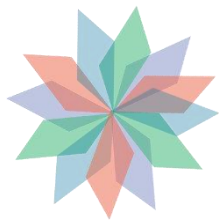
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Frequency Domain



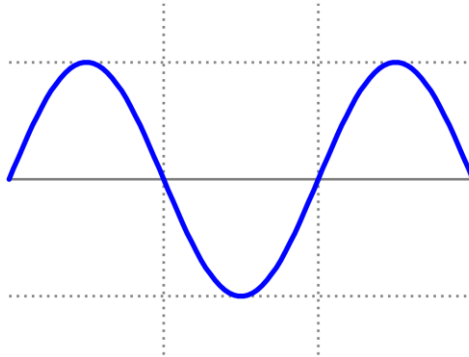


Fourier Series



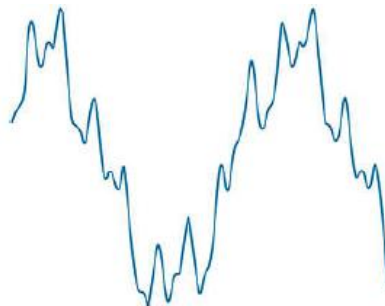
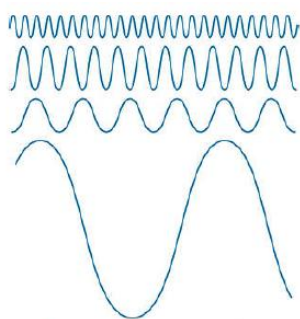
Fourier Series

- ▶ Contributed by Jean Baptiste Joseph Fourier, a French mathematician, in 1807
- ▶ Fourier series is a periodic function composed of sinusoids (sine waves)



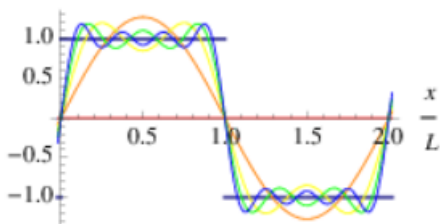
Fourier Series

- ▶ The idea is any functions can be approximated by a sum of sines and/or cosines of different frequencies, multiplied by a different coefficient.

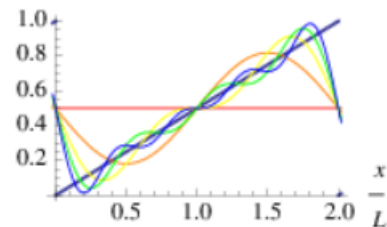


Fourier Series

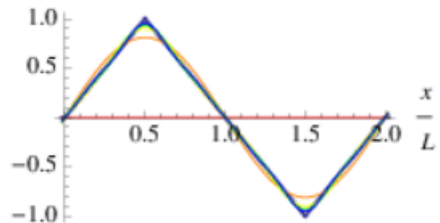
square wave



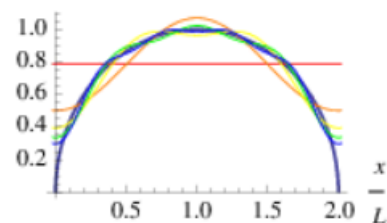
sawtooth wave



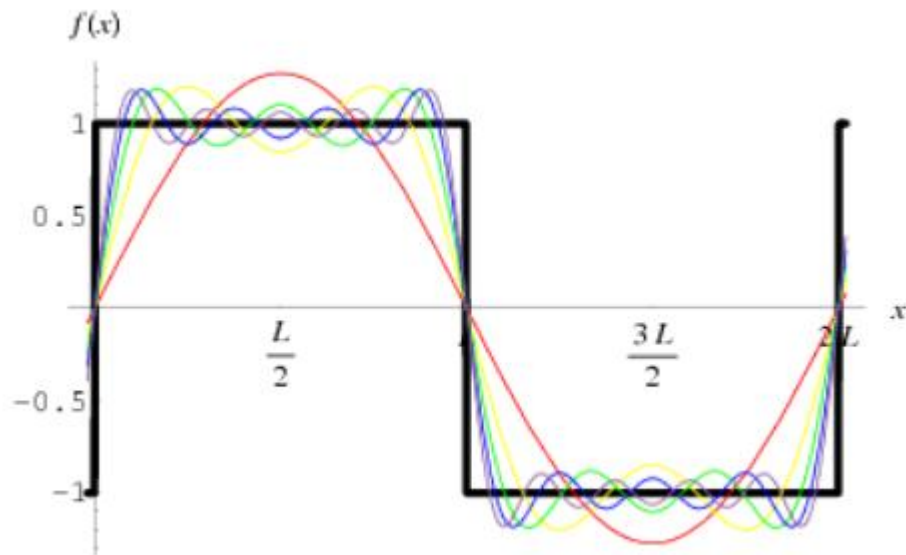
triangle wave



semicircle

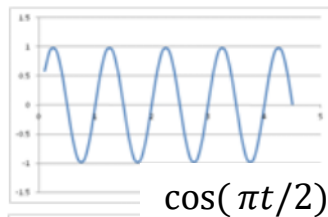


Fourier Series

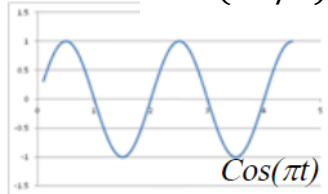




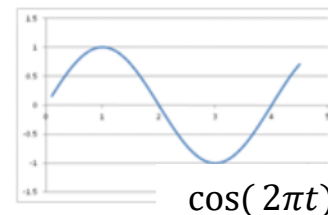
Fourier Series



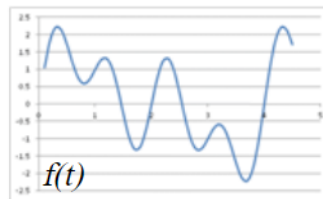
$\cos(\pi t/2)$



$\cos(\pi t)$



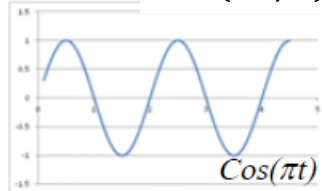
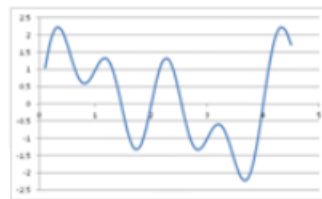
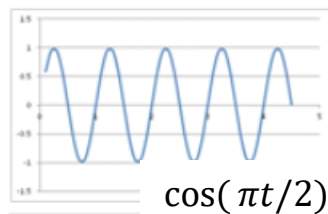
$\cos(2\pi t)$



$f(t)$

$$f(t) = \cos(\pi t/2) + \cos(\pi t) + \cos(2\pi t)$$

Fourier Series

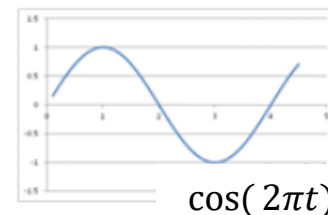


$$f(t) = \cos(\pi t/2) + \cos(\pi t) + \cos(2\pi t)$$

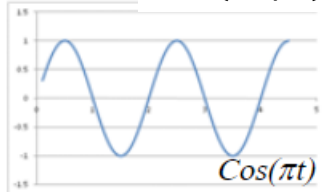
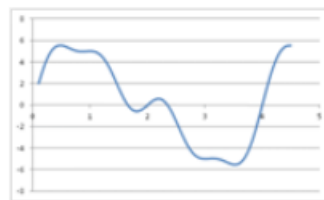
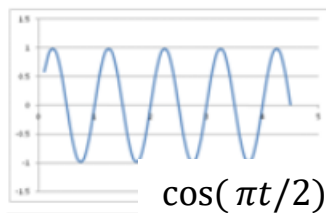
$$F_1 = 1 \quad u_1 = \pi/2$$

$$F_2 = 1 \quad u_2 = \pi$$

$$F_3 = 1 \quad u_3 = 2\pi$$



Fourier Series

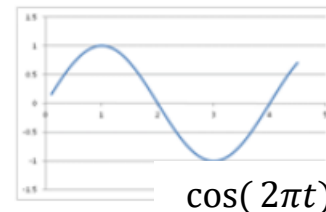


$$f(t) = 5\cos(\pi t/2) + 2\cos(\pi t) + \cos(2\pi t)$$

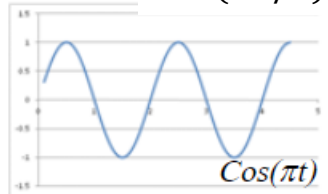
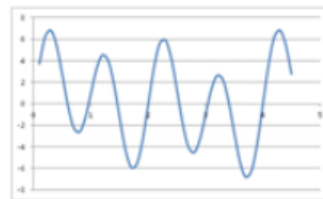
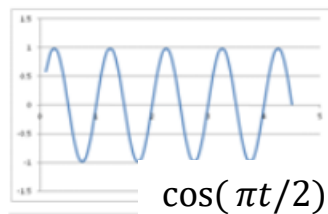
$$F_1 = 5 \quad u_1 = \pi/2$$

$$F_2 = 2 \quad u_2 = \pi$$

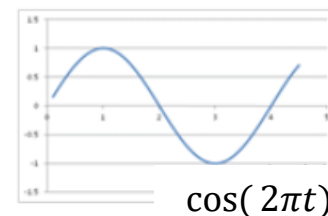
$$F_3 = 1 \quad u_3 = 2\pi$$



Fourier Series

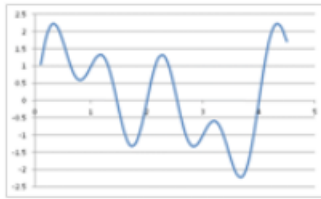


$$f(t) = \cos(\pi t/2) + 2\cos(\pi t) + 5\cos(2\pi t)$$

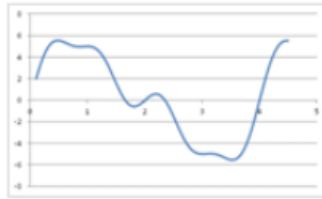


$$\begin{array}{ll} F_1 = 1 & u_1 = \pi/2 \\ F_2 = 2 & u_2 = \pi \\ F_3 = 5 & u_3 = 2\pi \end{array}$$

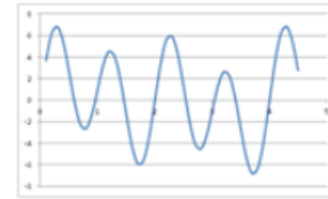
Fourier Series



$$\begin{aligned} F_1 &= 1 & u_1 &= \pi/2 \\ F_2 &= 1 & u_2 &= \pi \\ F_3 &= 1 & u_3 &= 2\pi \end{aligned}$$

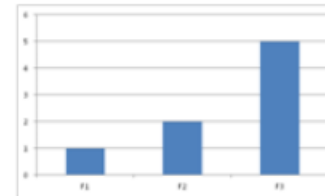
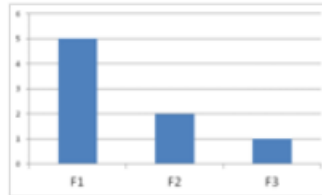
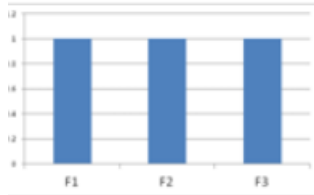


$$\begin{aligned} F_1 &= 5 & u_1 &= \pi/2 \\ F_2 &= 2 & u_2 &= \pi \\ F_3 &= 1 & u_3 &= 2\pi \end{aligned}$$



$$\begin{aligned} F_1 &= 1 & u_1 &= \pi/2 \\ F_2 &= 2 & u_2 &= \pi \\ F_3 &= 5 & u_3 &= 2\pi \end{aligned}$$

$$f(t) = F_1 * \cos(\pi t/2) + F_2 * \cos(\pi t) + F_3 * \cos(2\pi t)$$



**Power Spectrum or
Frequency Distribution**

Fourier Series

- Fourier series then can be written as

$$F(t) = \frac{a_0}{2} + \sum_{n=1}^N \left(a_n \cos \frac{2n\pi t}{T} + b_n \sin \frac{2n\pi t}{T} \right)$$

We will not go in a detail here. If you are interested to know how they calculate, visit:

<https://www.khanacademy.org/science/electrical-engineering/ee-signals#ee-fourier-series>

Fourier Series

Using Euler's formula,

$$e^{j\varphi} = \cos \varphi + j \sin \varphi ; j = \sqrt{-1} \text{ in electronic engineering}$$

Therefore, Fourier Series can be written in another form

$$F(t) = \sum_{n=-\infty}^{\infty} \left(c_n e^{j\frac{2\pi n}{T}t} \right)$$

Where

$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} F(t) e^{j\frac{2\pi n}{T}t} dt$$

Fourier Transform

Fourier Transform (FT) can be used to transform a function (often used in signal functions) into a Fourier series, in other words, frequency domain

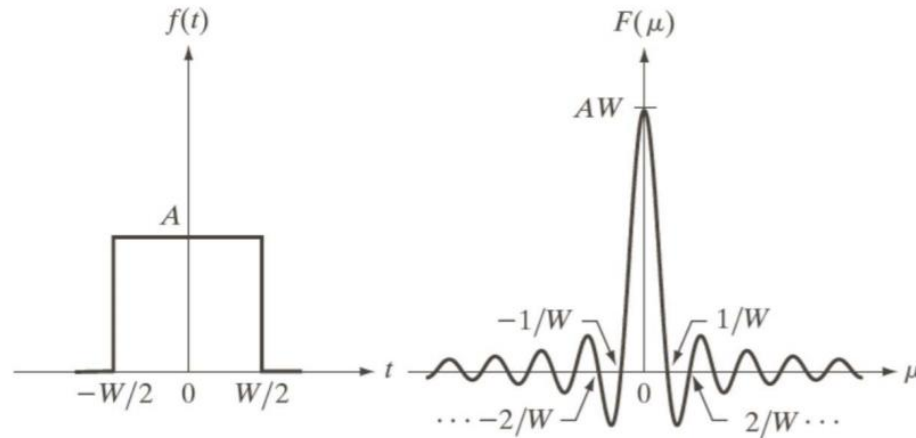
FT can be done by

$$F(\mu) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi\mu t} dt$$

Where μ is a continuous variable

Fourier Transform

For example, a simple function $f(t) = A$ where t is between $-w/2$ and $w/2$ can be transformed as follow



Fourier Transform

Conversely, a Fourier series can be transform back using inverse Fourier transform, written as

$$f(t) = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} dt$$

Where μ is a continuous variable



Two-Dimensional Fourier Transform

Two-dimensional Fourier Transform

- ▶ One-dimensional Fourier transform is normally used in time-series (temporal domain)
- ▶ However, in other domain such as spatial domain, only one dimensional is not enough

Two-dimensional Fourier Transform

Fourier transform in two-dimensional is as follow

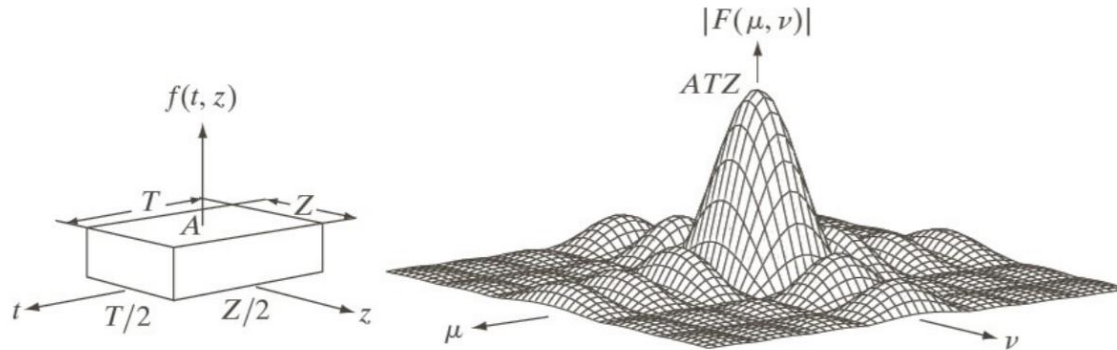
$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy,$$

While its inverse is

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} du dv$$

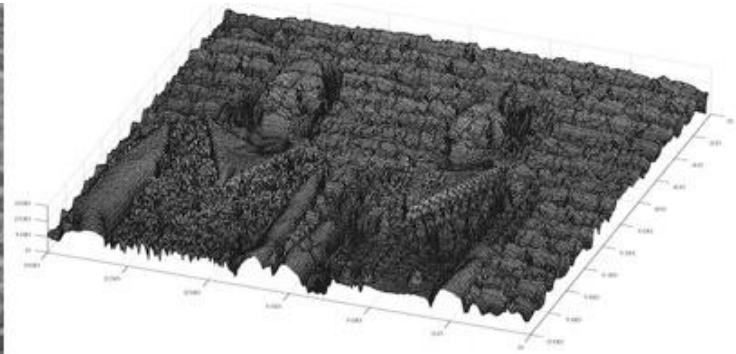
Two-dimensional Fourier Transform

For example, a simple function $f(t, z) = A$ where t is between $-T/2$ and $T/2$, z is between $-Z/2$ and $Z/2$ can be transformed as follow



Fourier Transform in digital images

- ▶ An image is a 2D (spatial and intensity) function
- ▶ The image on the right represents the function $f(x, y)$ of intensities (grayscale levels) of the image on the left



Fourier Transform in digital images

- ▶ However, as we all know that digital image is a discrete function, not a continuous function
- ▶ Therefore, the Fourier transform cannot be used with digital image!



Discrete Fourier Transform (DFT)

Discrete Fourier Transform

- Discrete Fourier Transform (DFT) of a digital image with a size $M \times N$ is

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

While its inverse is

$$f(x, y) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} F(u, v) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

Discrete Fourier Transform

- ▶ Using Euler's formula

$$e^{j\varphi} = \cos \varphi + j \sin \varphi$$

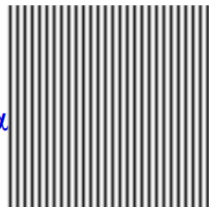
DFT can be rewritten as

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \left[\cos \left(2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right) \right) - j \sin \left(2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right) \right) \right]$$

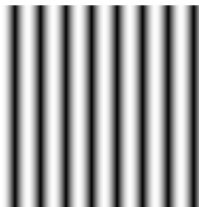
$f(x, y)$



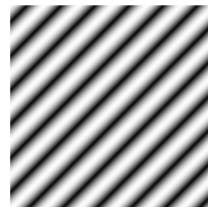
$= \alpha$



$+ \beta$



$+ \gamma$



$+ \dots$

Discrete Fourier Transform

► Forward DFT Pseudo Code (1D)

For all transform points ($u = [0, M-1]$) do

$F(u).real = 0$

$F(u).im = 0$

For all discrete data points ($x = [0, M-1]$) do

$F(u).real += F(x) \cos(\frac{2\pi ux}{M})$

$F(u).im -= F(x) \sin(\frac{2\pi ux}{M})$

End For

$F(u).real = sum$

$F(u).im = sum$

End For

Discrete Fourier Transform

► Inverse DFT Pseudo Code (1D)

For all transform points ($x = [0, M-1]$) do

$F(u).real = 0$

$F(u).im = 0$

For all discrete data points ($u = [0, M-1]$) do

$F(u).real += F(x) \cos(\frac{2\pi ux}{M})$

$F(u).im += F(x) \sin(\frac{2\pi ux}{M})$

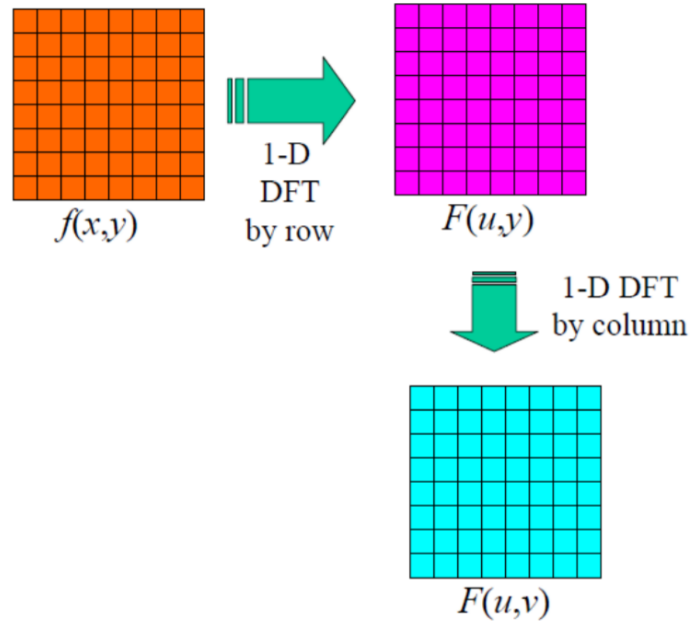
End For

$F(u).real = \frac{sum}{M}$

$F(u).im = \frac{sum}{M}$

End For

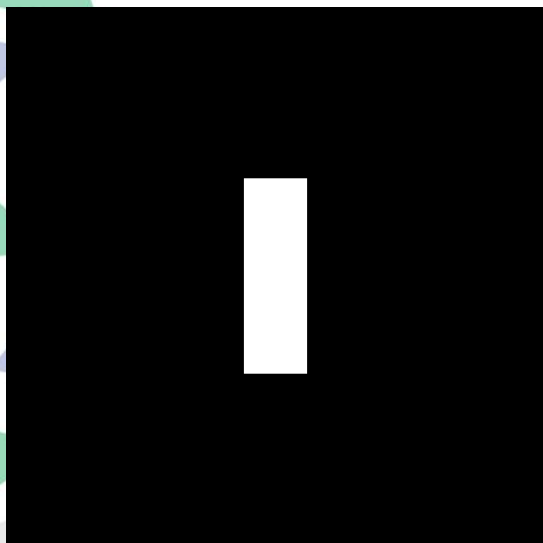
Discrete Fourier Transform



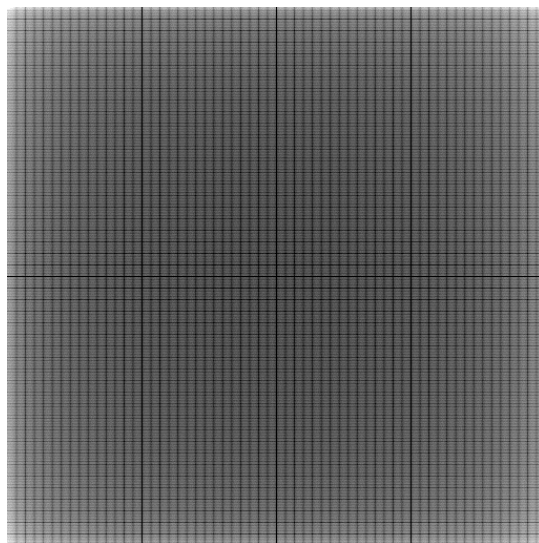
Discrete Fourier Transform

- ▶ In the end, $F(u)$ is a complex number $\text{Re}(u) + j \text{Im}(u)$
- ▶ Therefore, when showing frequency domain of an image, there will be 2 images to represent magnitude and phase
 - ▶ $Magnitude = \sqrt{\text{Re}^2(u) + \text{Im}^2(u)}$
 - ▶ $Phase = \theta(u) = \tan^{-1}\left[\frac{\text{Im}(u)}{\text{Re}(u)}\right]$
- ▶ Normally, we usually display only Magnitude part for purpose of viewing

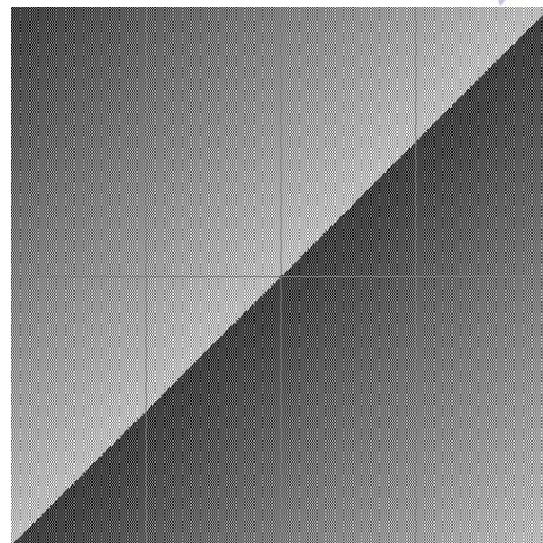
Discrete Fourier Transform



Original Image



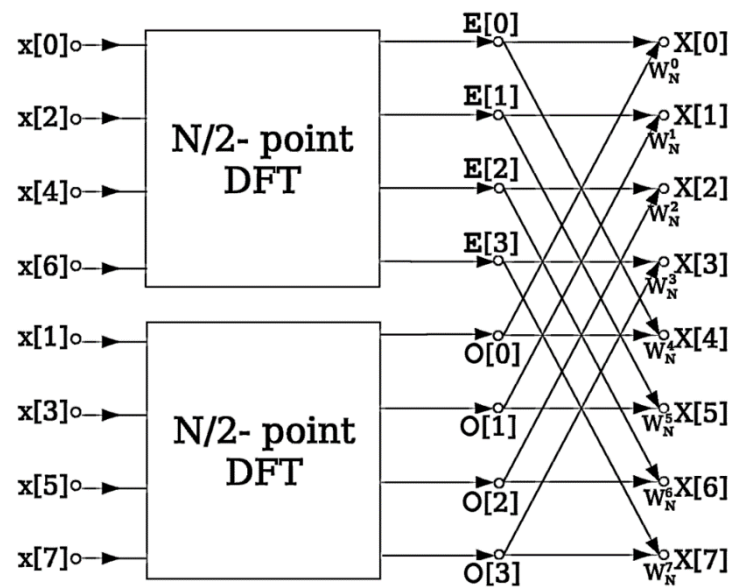
Spectrum Magnitude



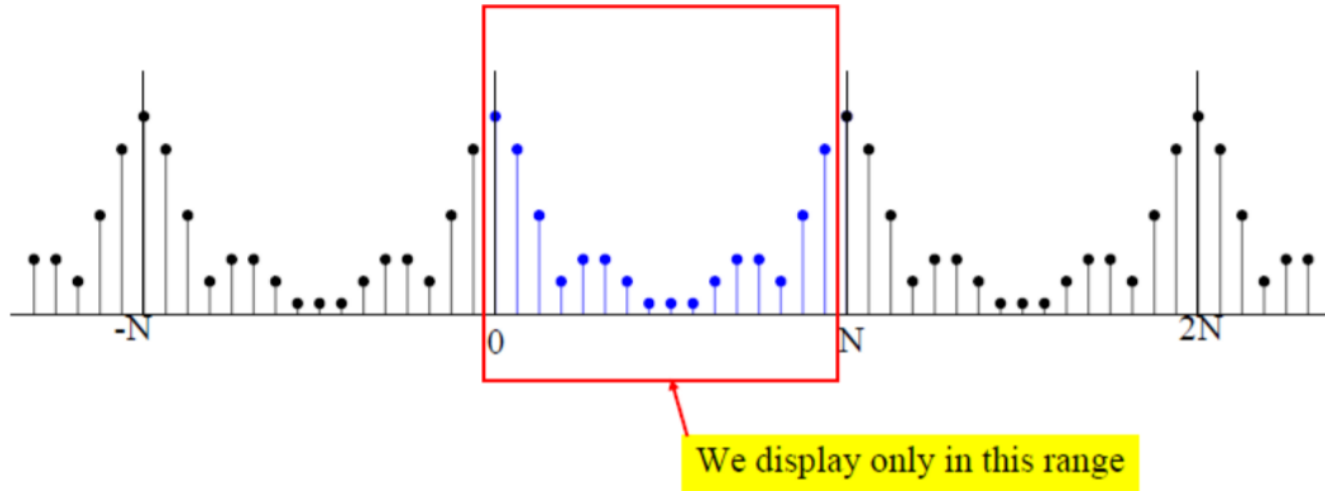
Spectrum Phase

Discrete Fourier Transform

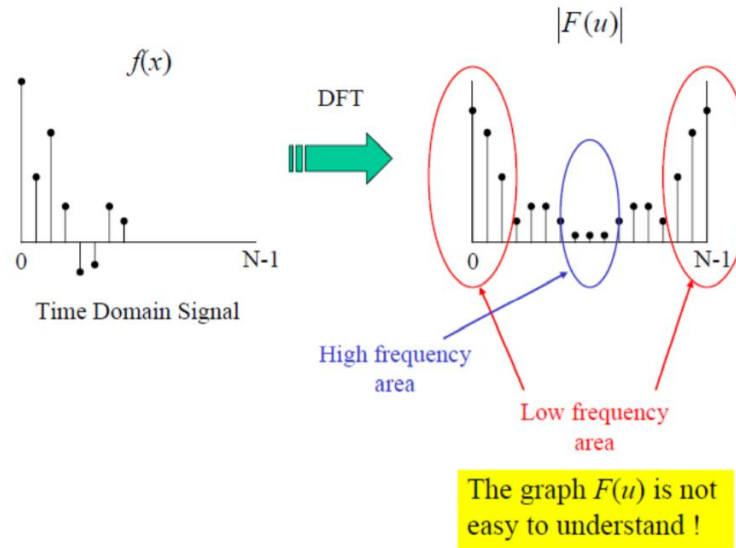
- ▶ Fast Fourier Transform (FFT)
 - ▶ DFT's complexity is $O(n^2)$ which is too slow in practical for computing an image
 - ▶ FFT is widely used in practical which its complexity is $O(N \log N)$
 - ▶ There are several algorithms. One well-known algorithm is **Cooley–Tukey FFT algorithm (DIT-FFT)**
 - ▶ **Continued in the lab**



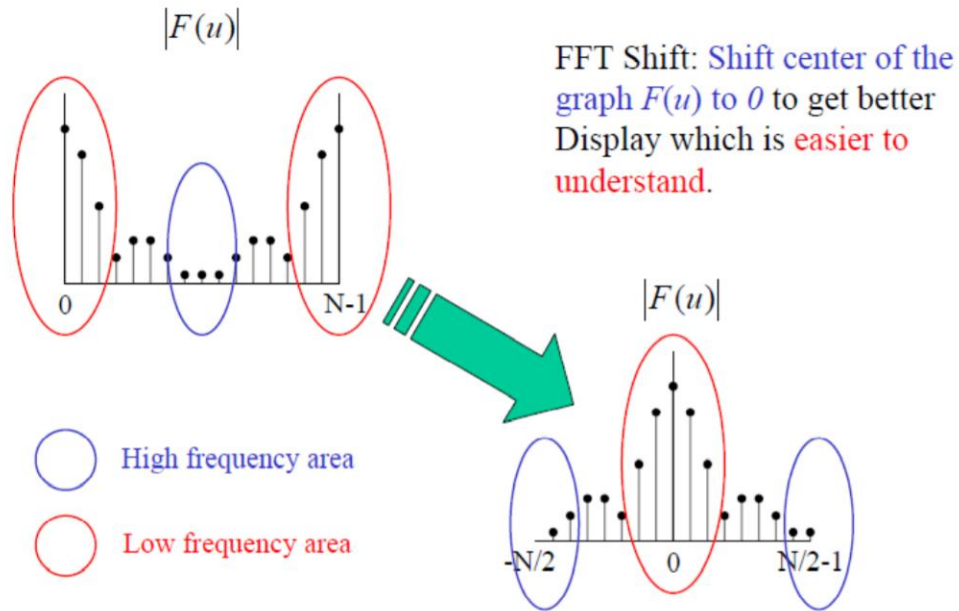
Discrete Fourier Transform



Discrete Fourier Transform

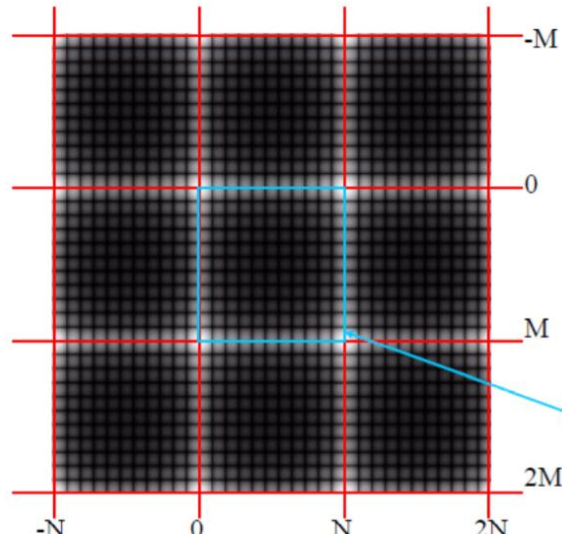
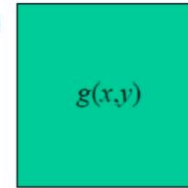


Discrete Fourier Transform



Discrete Fourier Transform

2-D DFT:
$$F(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$



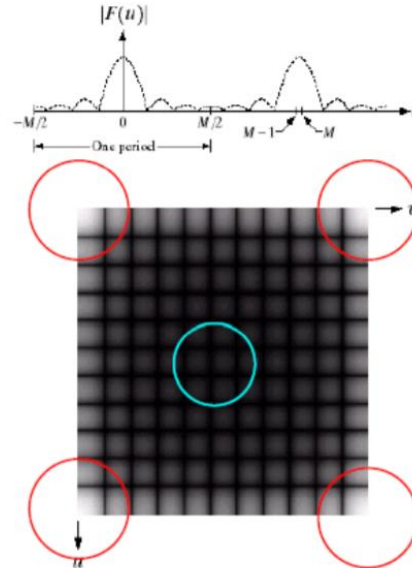
For an image of size $N \times M$ pixels, its 2-D DFT repeats itself every N points in x -direction and every M points in y -direction.

We display only in this range

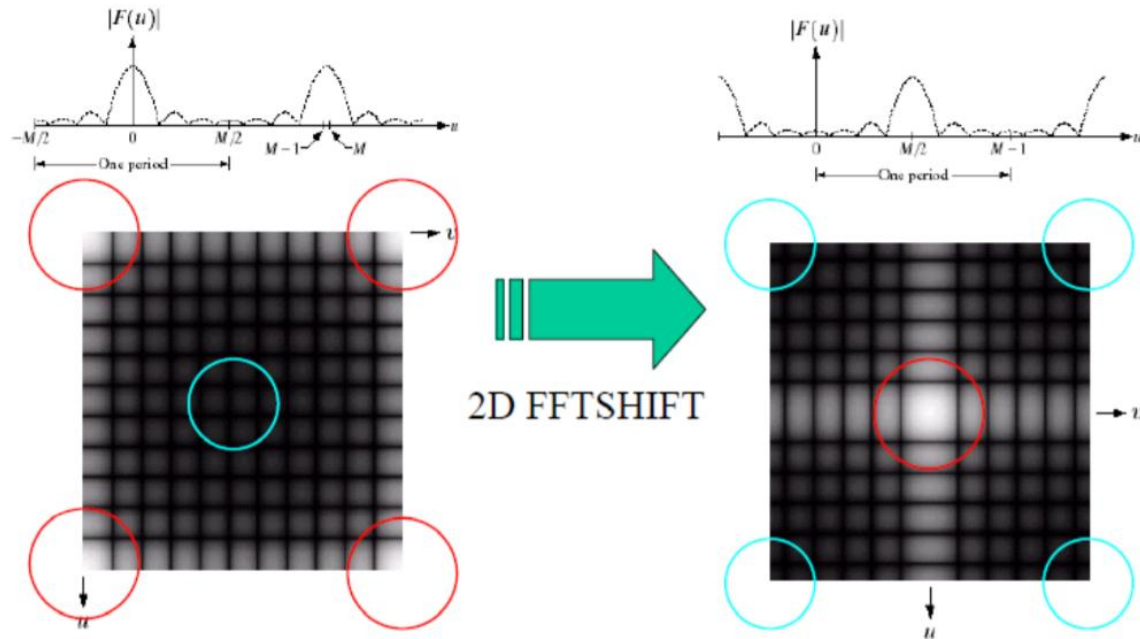
Discrete Fourier Transform

$F(u,v)$ has low frequency areas at corners of the image while high frequency areas are at the center of the image which is inconvenient to interpret.

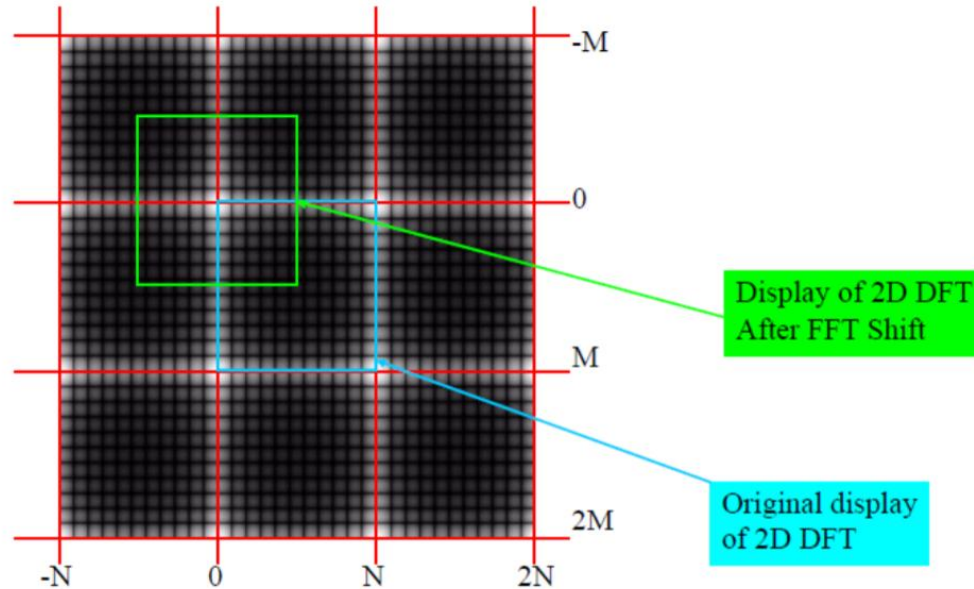
- High frequency area
- Low frequency area



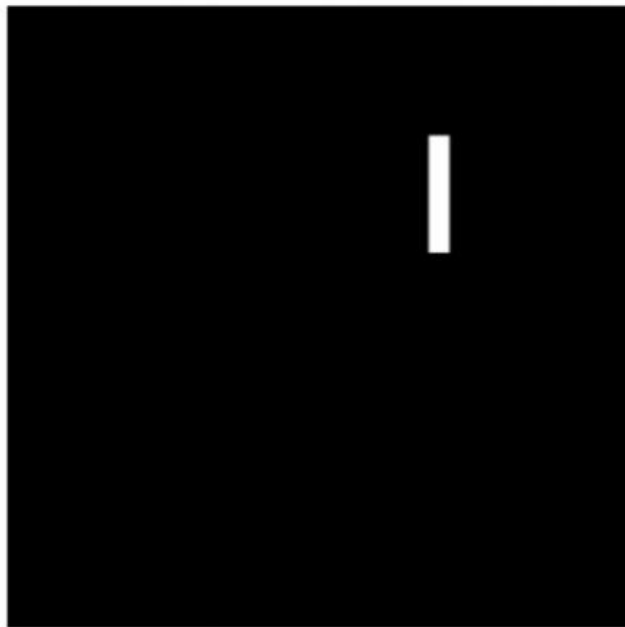
Discrete Fourier Transform



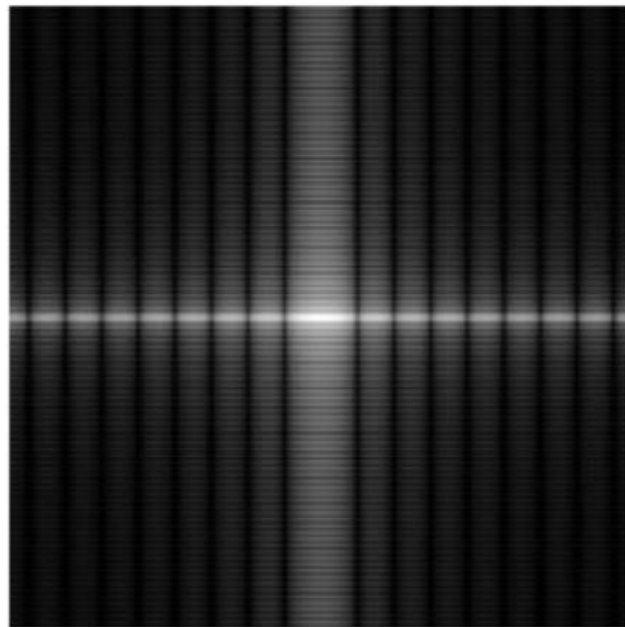
Discrete Fourier Transform



Discrete Fourier Transform

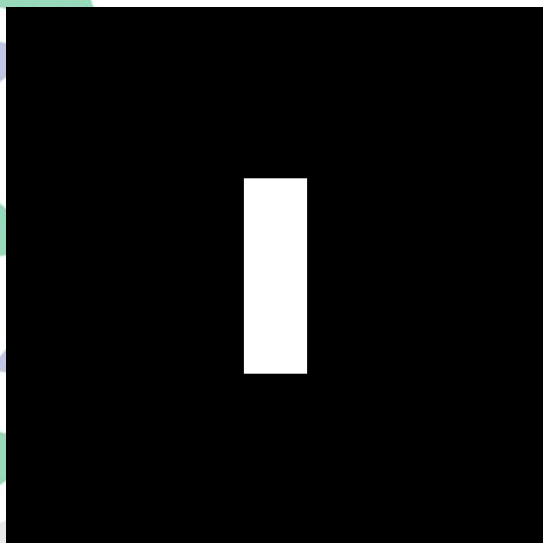


Original Image

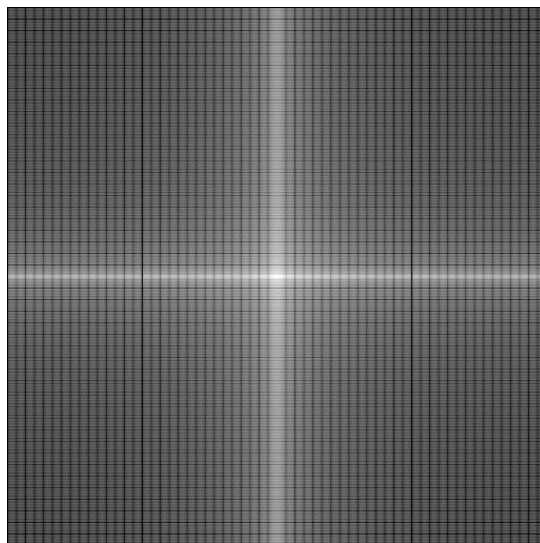


Spectrum Magnitude

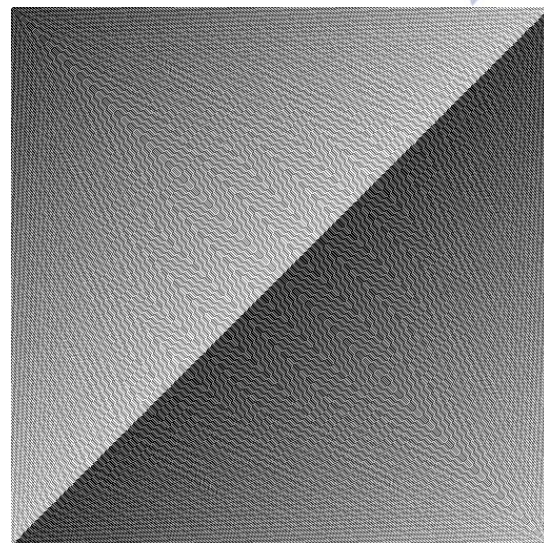
Discrete Fourier Transform



Original Image

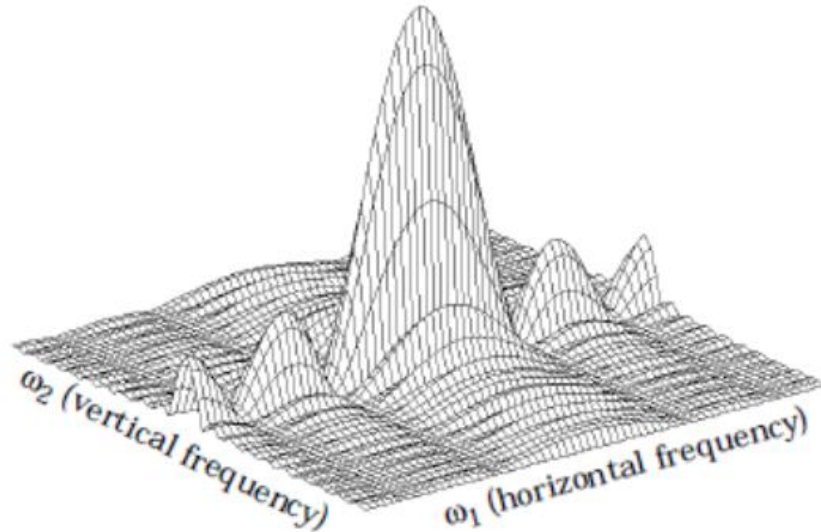
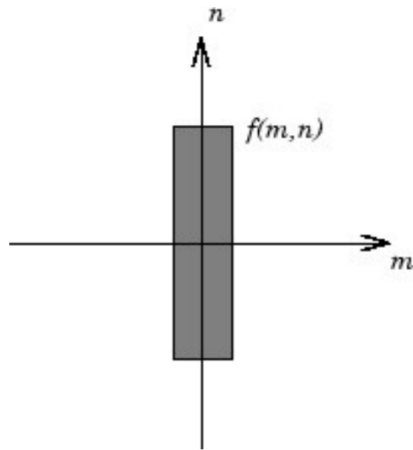


Spectrum Magnitude

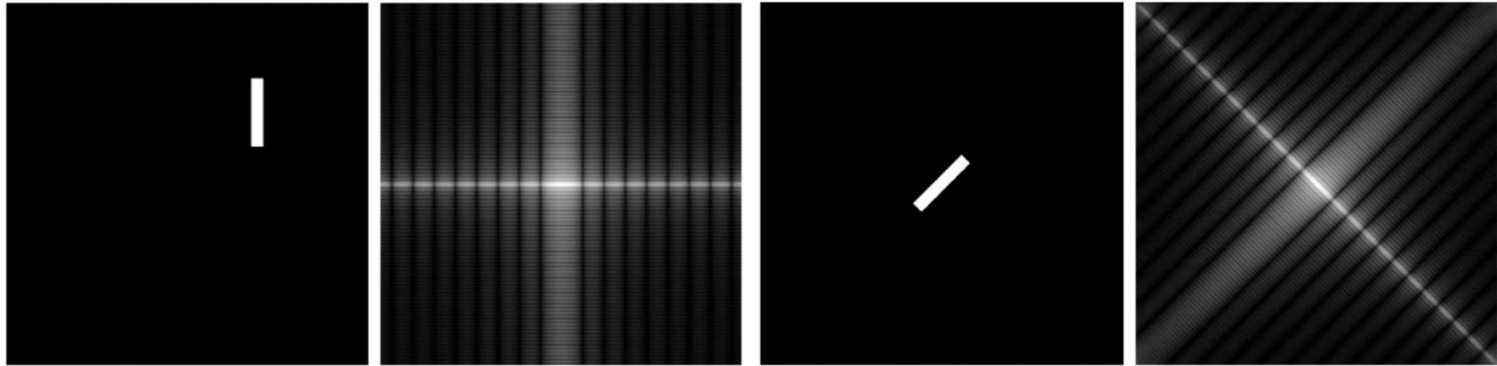


Spectrum Phase

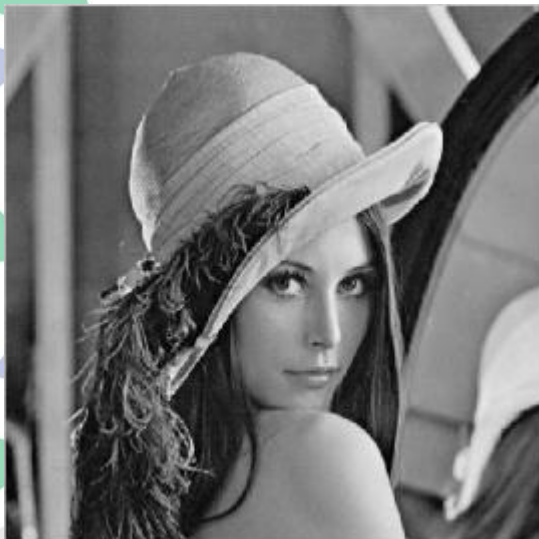
Discrete Fourier Transform



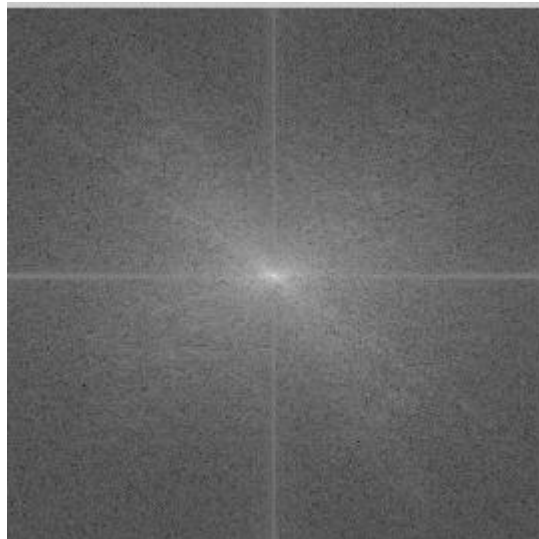
Discrete Fourier Transform



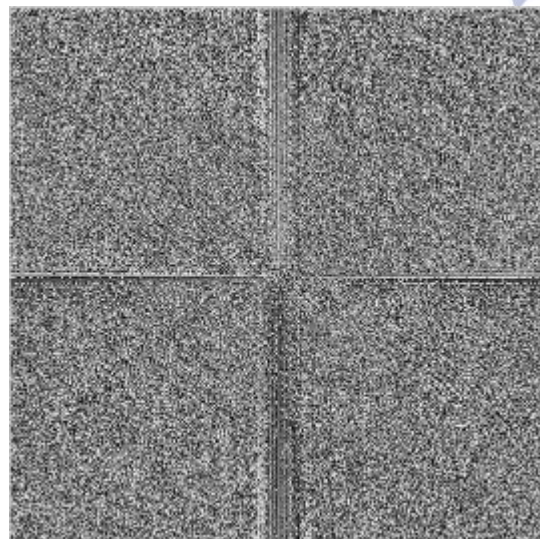
Discrete Fourier Transform



Original Image

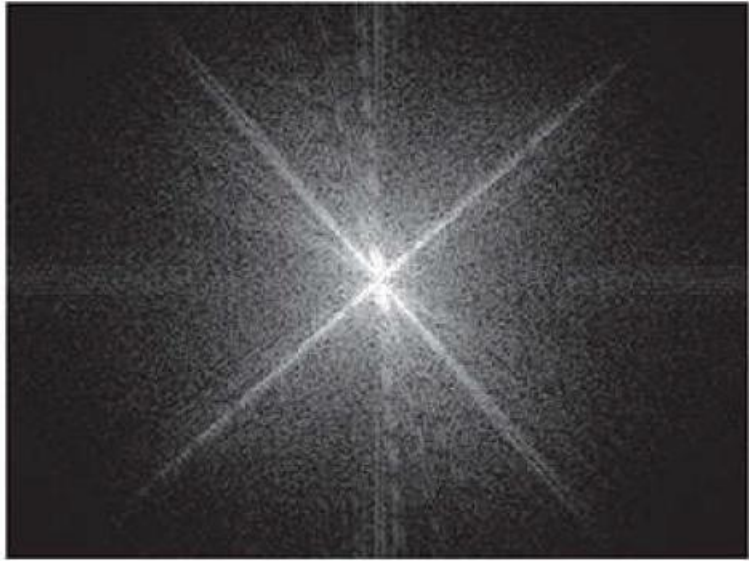
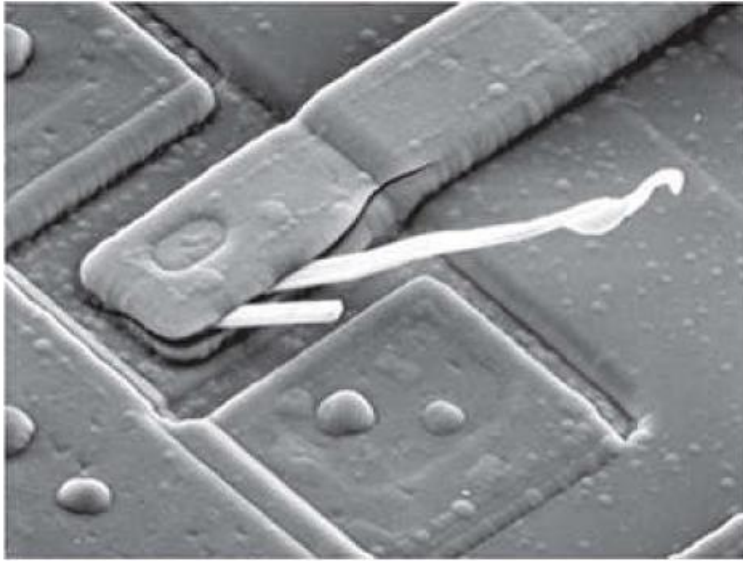


Spectrum Magnitude



Spectrum Phase

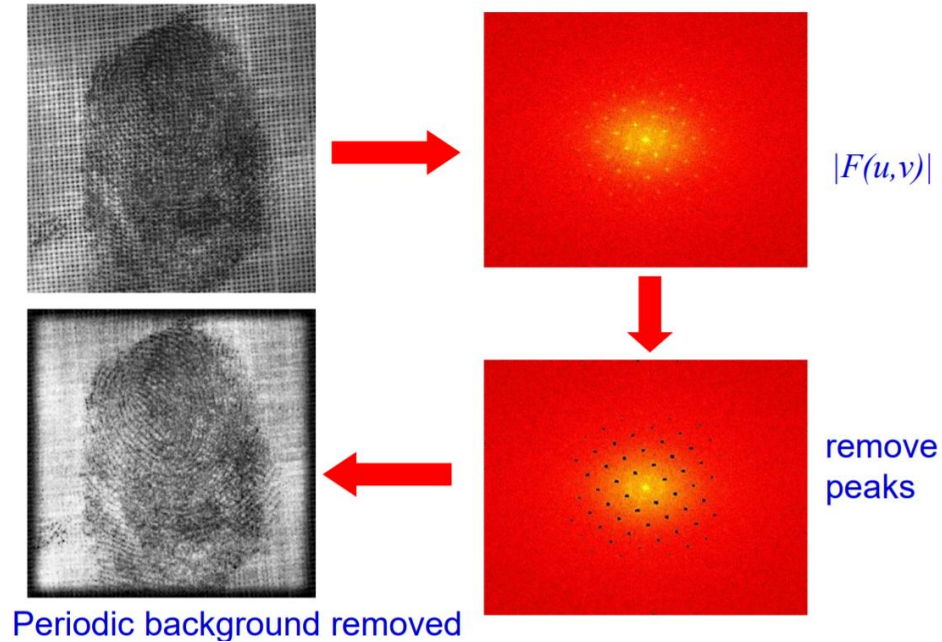
Discrete Fourier Transform





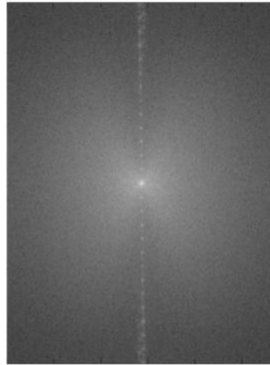
DFT Applications

Discrete Fourier Transform

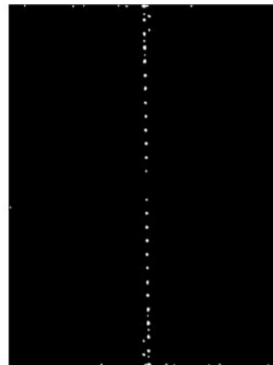


Discrete Fourier Transform

Lunar orbital image (1966)



$|F(u,v)|$



remove
peaks



join lines
removed



Filtering in Frequency Domain

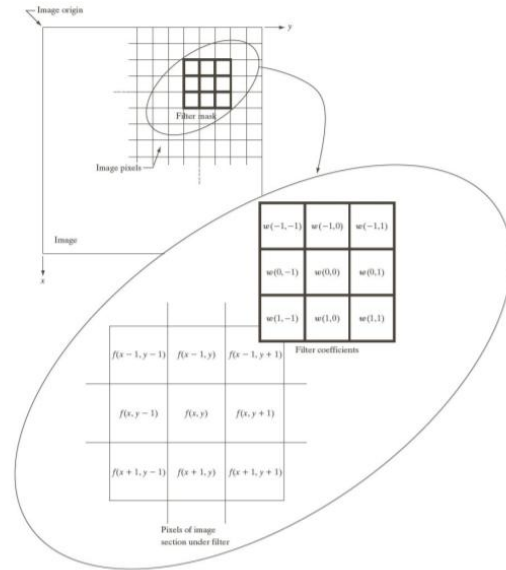
Filtering in Frequency Domain

- Recalling filtering in spatial domain

$$\begin{aligned} g(x, y) &= w(-1, -1)f(x - 1, y - 1) \\ &+ w(-1, 0)f(x - 1, y) + \dots + w(0, 0)f(x, y) \\ &+ \dots w(1, 1)f(x + 1, y + 1) \end{aligned}$$

Or

$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t)f(x + s, y + t)$$



Filtering in Frequency Domain

- ▶ Linear filtering operations in spatial domain can be carried out by simple multiplications in the frequency domain

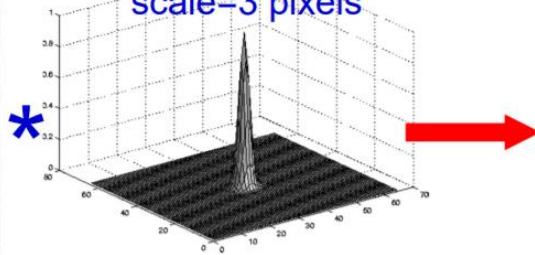
$$f(x, y) * h(x, y) \Leftrightarrow F(u, v)H(u, v)$$

Space convolution = frequency multiplication

$f(x,y)$



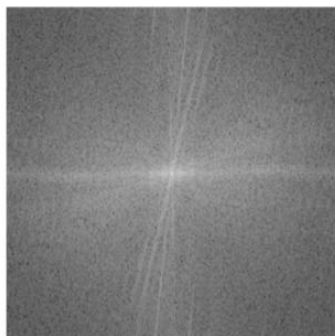
Gaussian
scale=3 pixels



$g(x,y)$

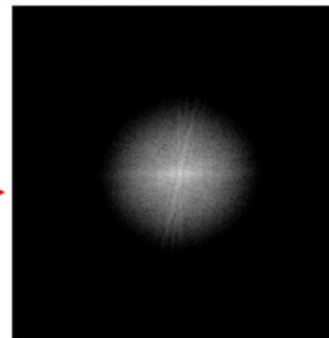
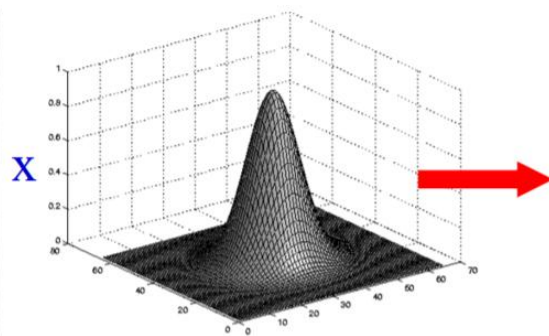


Fourier transform

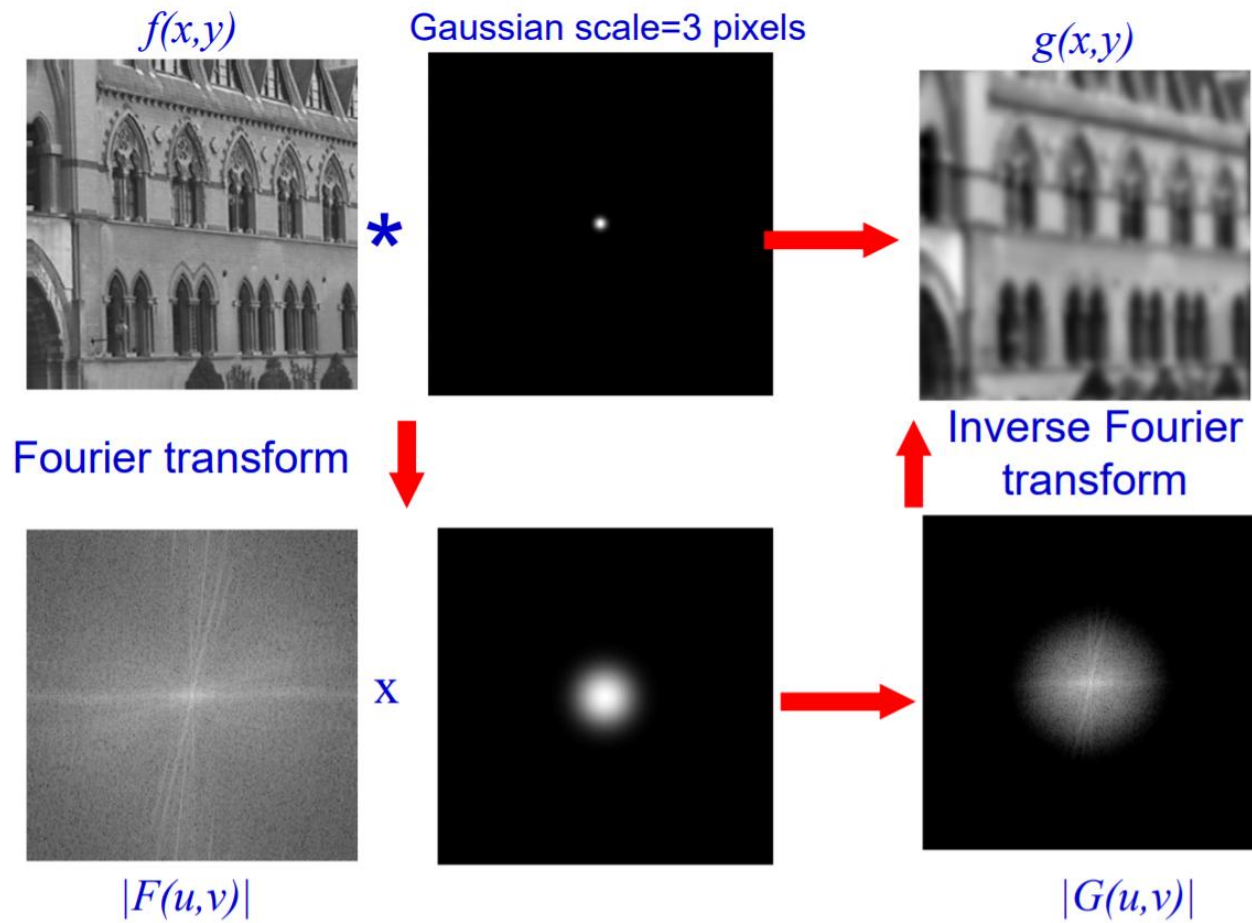


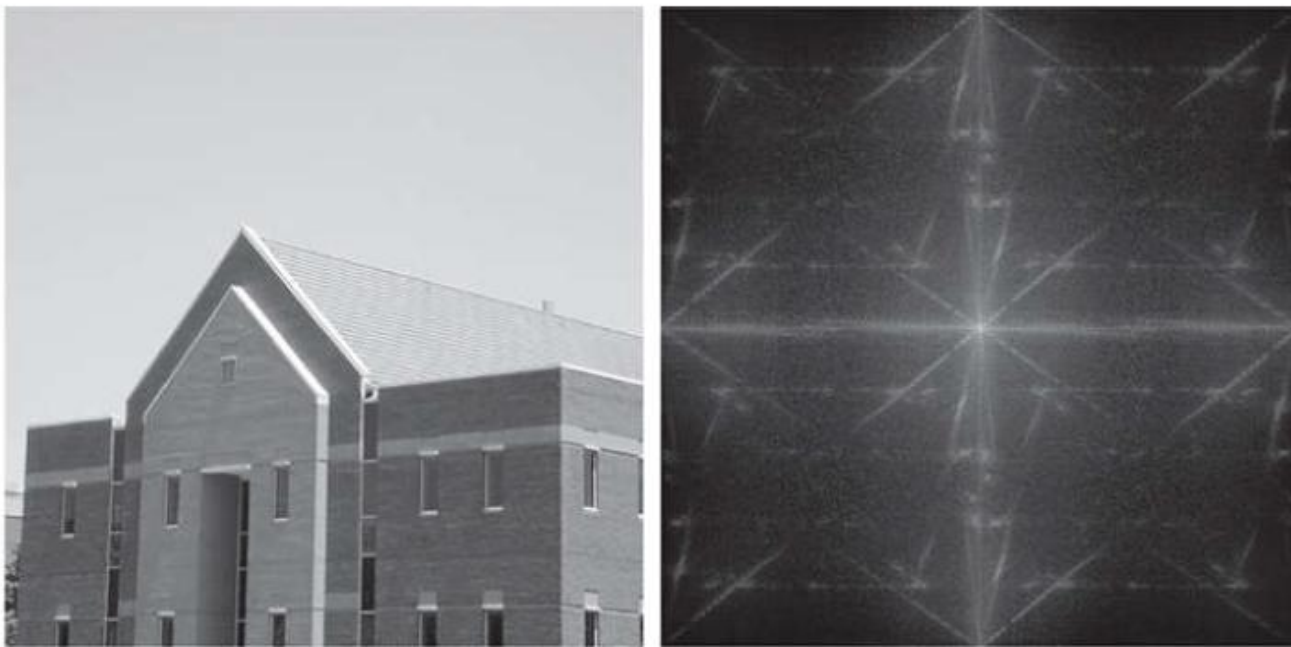
$|F(u,v)|$

Inverse Fourier
transform

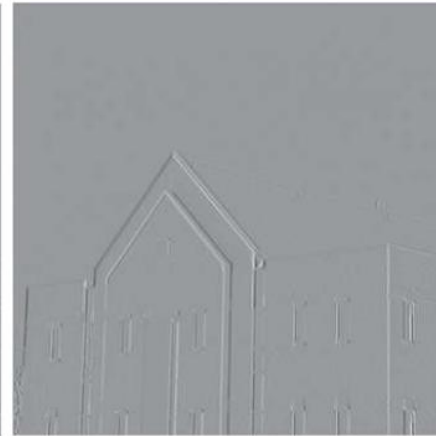
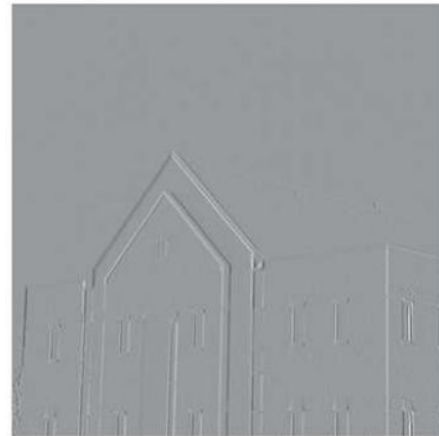
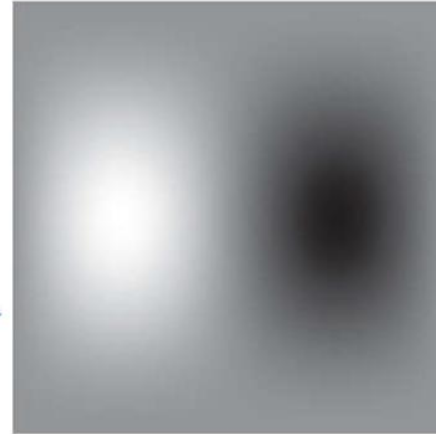
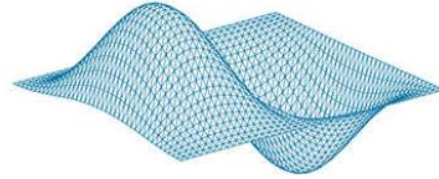


$|G(u,v)|$





-1	0	1
-2	0	2
-1	0	1



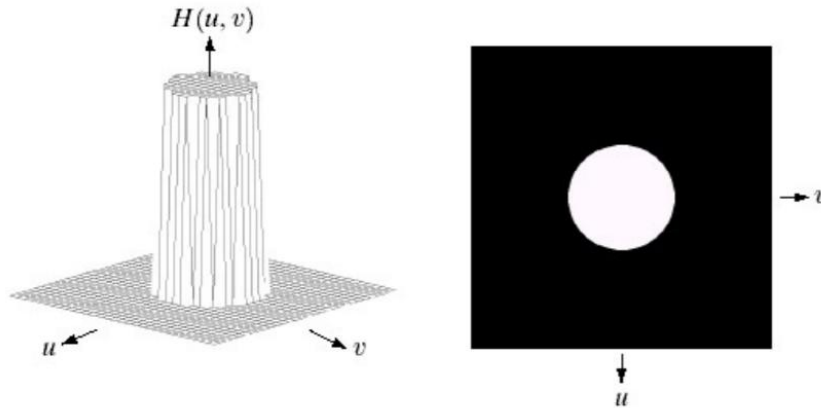
Low vs High Frequency

- ▶ Low frequency is the area where the intensities of the image are changed slowly e.g. background colour, etc.
- ▶ High frequency is the area where the intensities of the image are changed quickly e.g. noise, edge, etc.

Lowpass vs Highpass Filter

- ▶ Lowpass filter is a filter which allows low frequencies to get through and reduce the high frequency
- ▶ Highpass filter is a filter which allows high frequencies to get through and reduce the low frequency

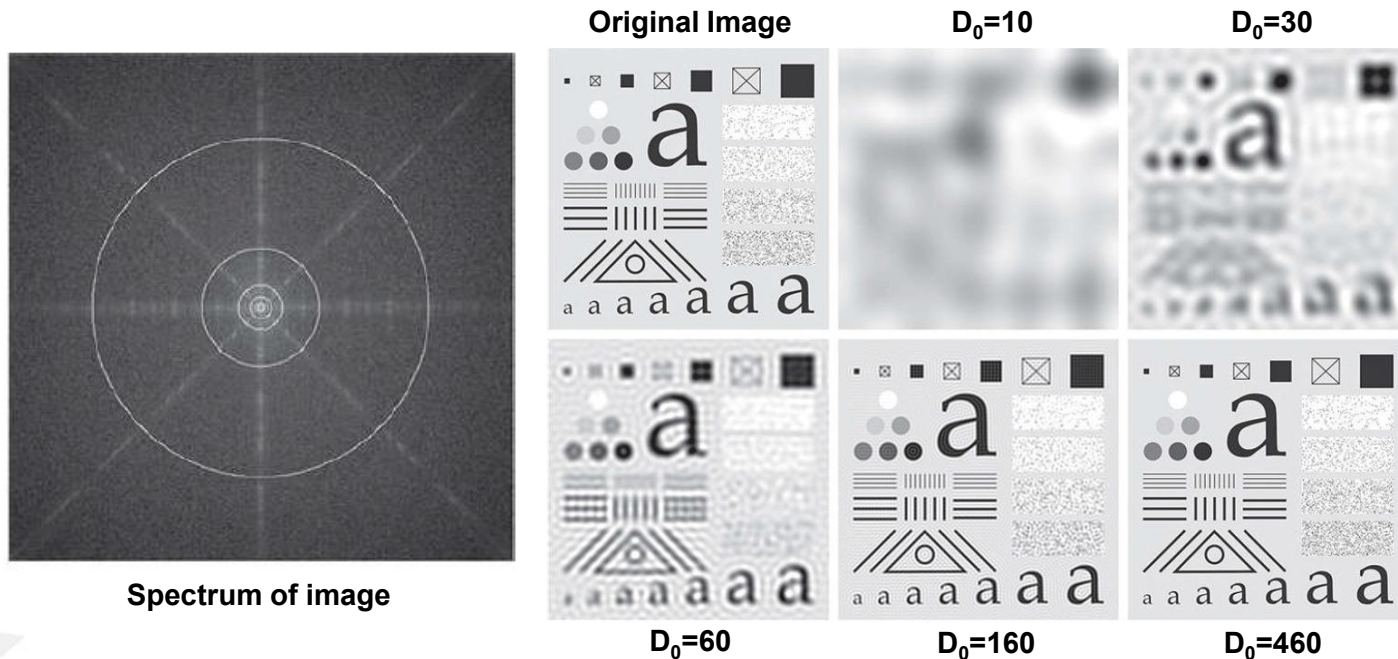
Ideal Lowpass Filter (ILPF)



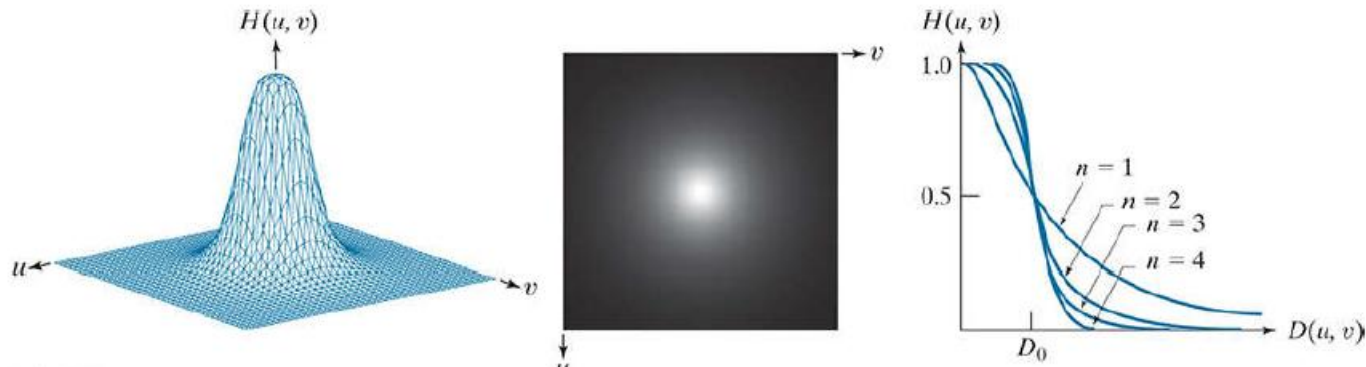
$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$$

$$D(u, v) = \sqrt{\left(u - \frac{M}{2}\right)^2 + \left(v - \frac{N}{2}\right)^2}$$

Ideal Lowpass Filter (ILPF)

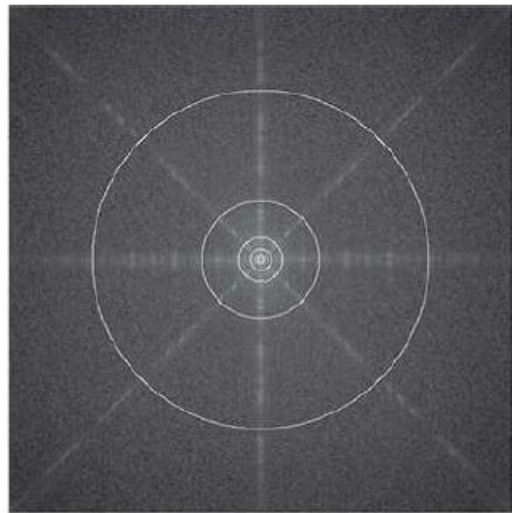


Butterworth Lowpass Filter (BLPF)



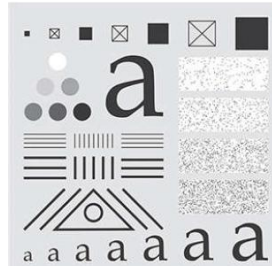
$$H(u, v) = \frac{1}{1 + [D(u, v)/D_0]^{2n}}$$

Butterworth Lowpass Filter (BLPF)



Spectrum of image

Original Image



$D_0=5$

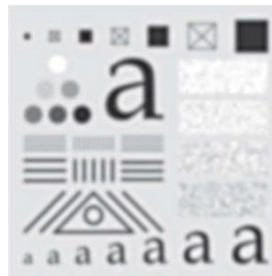


$D_0=15$

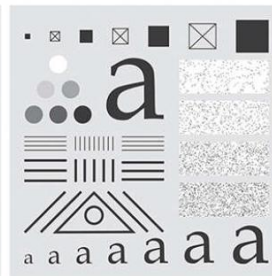


$n=2.25$

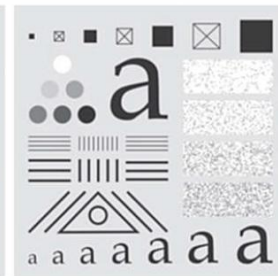
$D_0=30$



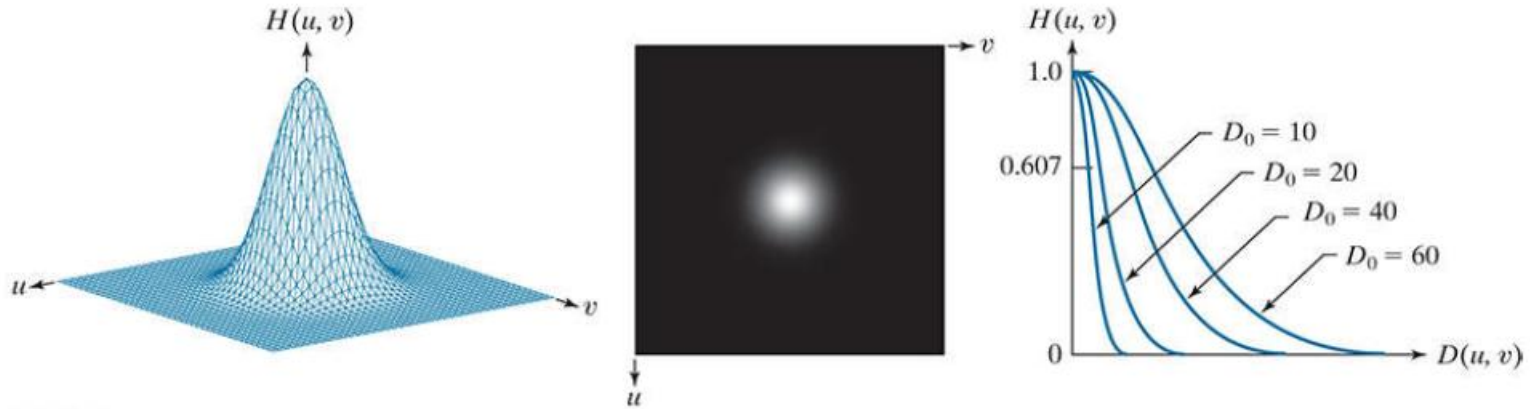
$D_0=80$



$D_0=230$

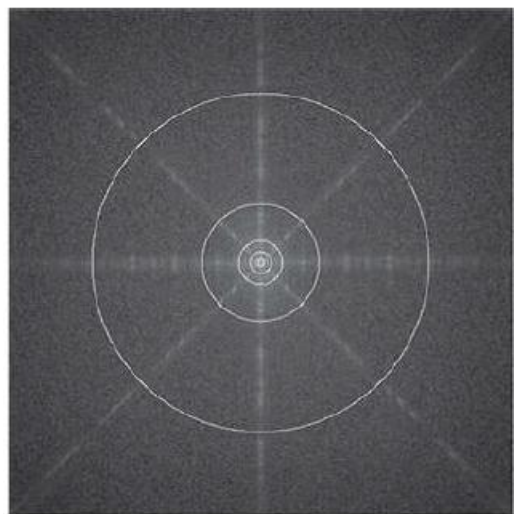


Gaussian Lowpass Filter (GLPF)



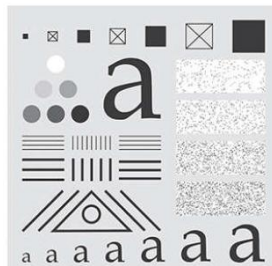
$$H(u, v) = e^{-D^2(u, v) / 2D_0^2}$$

Gaussian Lowpass Filter (GLPF)



Spectrum of image

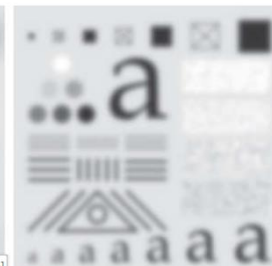
Original Image



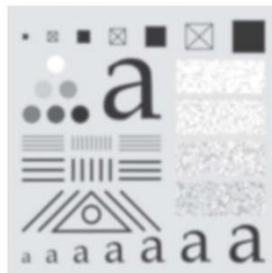
$D_0=5$



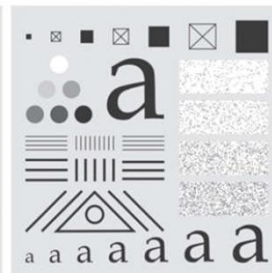
$D_0=15$



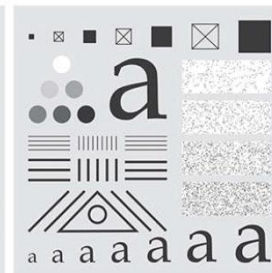
$D_0=30$



$D_0=80$



$D_0=230$



Gaussian Lowpass Filter (GLPF)

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



ea

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



ea

Gaussian Lowpass Filter (GLPF)



Original Image

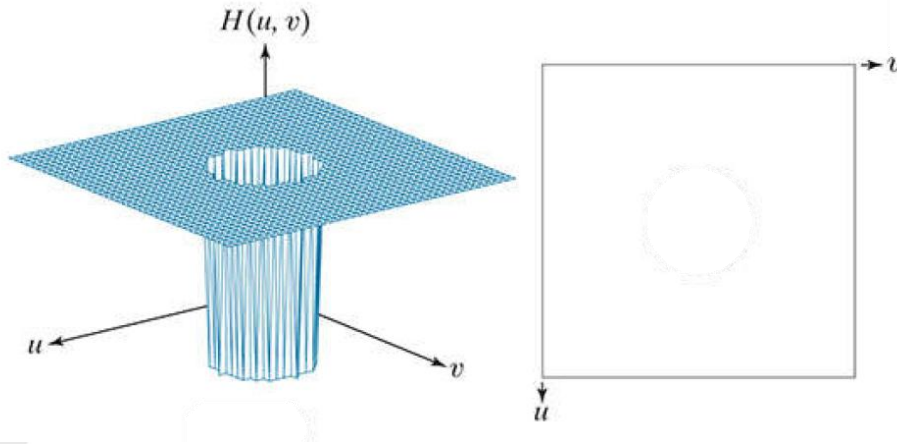
$D_0=150$

$D_0=130$

Lowpass vs Highpass Filter

$$H_{\text{HP}}(u, v) = 1 - H_{\text{LP}}(u, v)$$

Ideal Highpass Filter (IHPF)



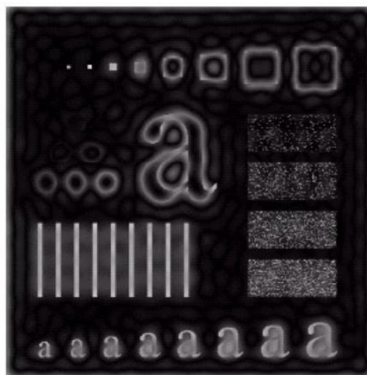
$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$$

$$D(u, v) = \sqrt{\left(u - \frac{M}{2}\right)^2 + \left(v - \frac{N}{2}\right)^2}$$

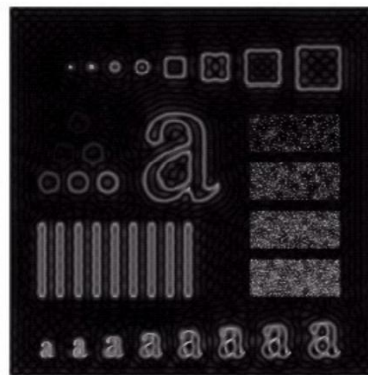
Ideal Highpass Filter (IHPF)



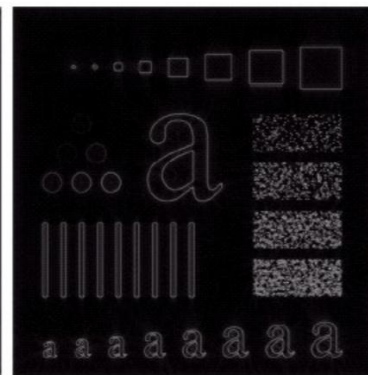
Original Image



$D_0=15$

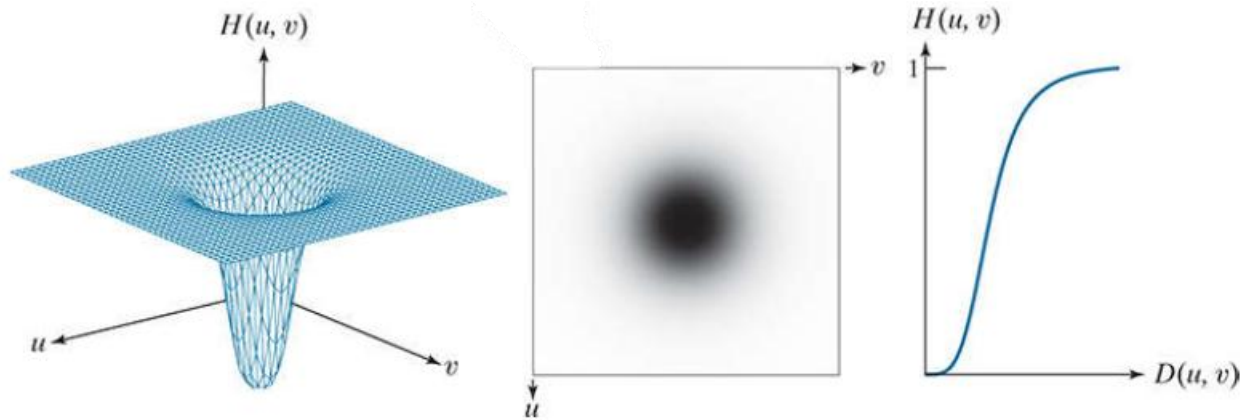


$D_0=30$



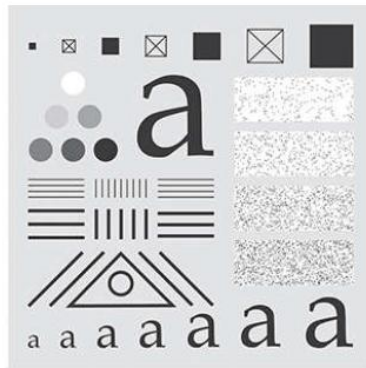
$D_0=80$

Butterworth Highpass Filter (BHPF)

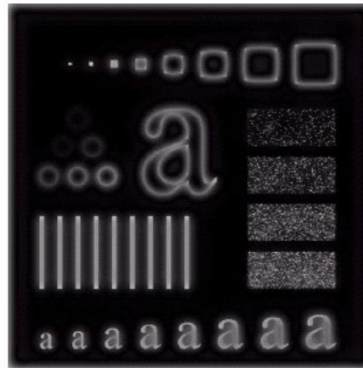


$$H(u, v) = \frac{1}{1 + [D_0/D(u, v)]^{2n}}$$

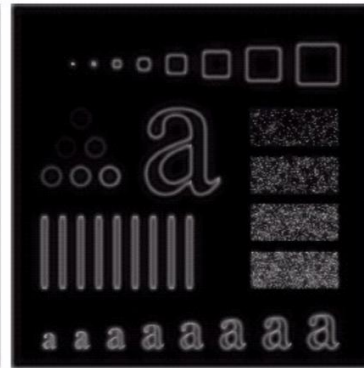
Butterworth Highpass Filter (BHPF)



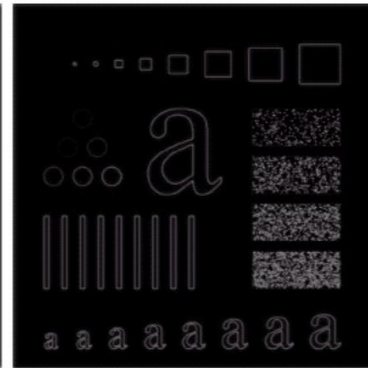
Original Image



$D_0=15$

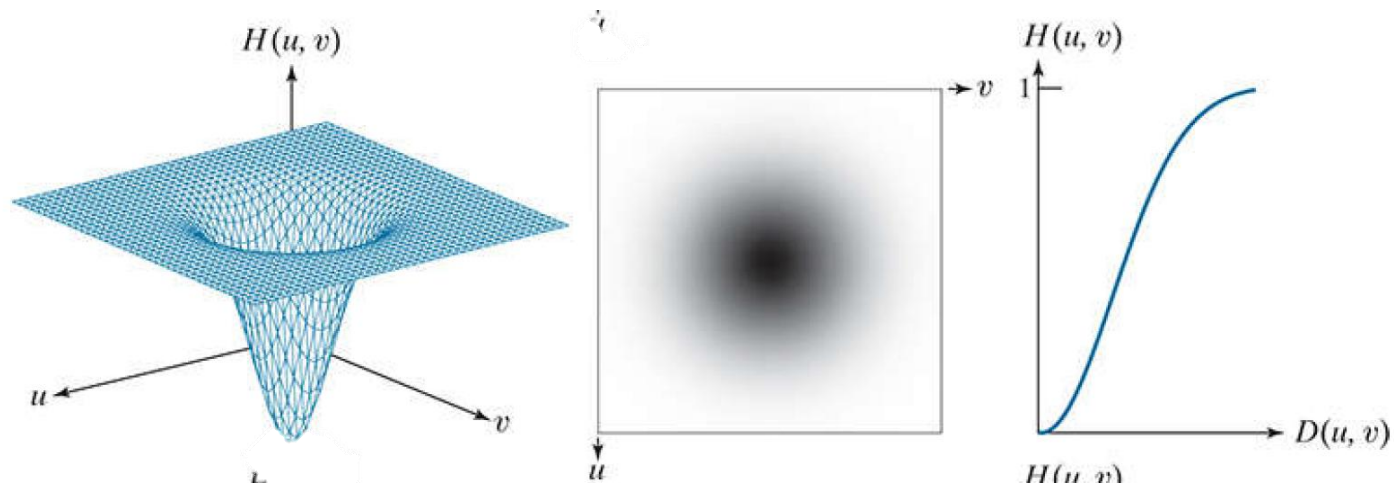


$D_0=30$



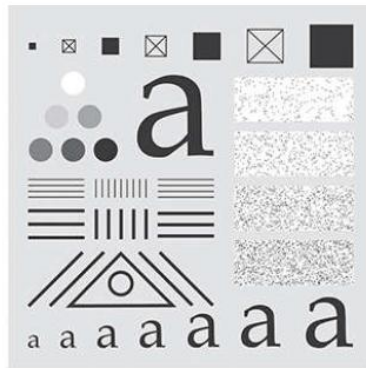
$D_0=80$

Gaussian Highpass Filter (GHPF)

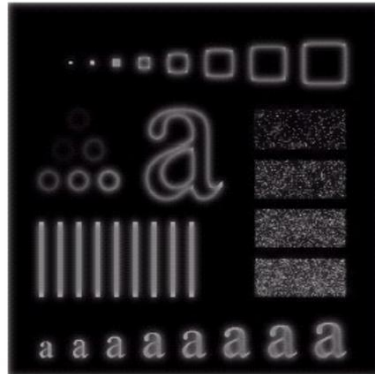


$$H(u, v) = 1 - e^{-D^2(u, v)/2D_0^2}$$

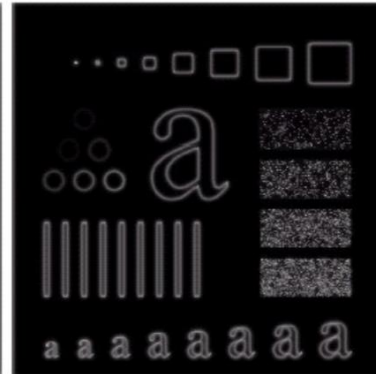
Gaussian Highpass Filter (GHPF)



Original Image



$D_0=15$



$D_0=30$



$D_0=80$



End of Week 5